Comparing PID and Reinforcement-Learning (PPO) Controllers for Spacecraft Attitude Control

A simulation study

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Code and data: github.com/yasinkh81b-ui/ppo-vs-pid-spacecraft-attitude

Abstract. Reinforcement learning (RL) is increasingly proposed for spacecraft attitude control, but claims of superiority are often made against weakly tuned classical baselines and from single training runs. This study compares a PD controller, a PID controller with anti-windup, and two Proximal Policy Optimization (PPO) variants on a simulated rigid spacecraft with actuator saturation and an unknown constant disturbance torque. Classical gains were tuned by grid search on 10 separate tuning scenarios; all controllers were tested on 100 unseen random scenarios drawn from the same distribution used to train the RL agents, and each PPO variant was trained five times with independent random seeds. The tuned PID achieved the lowest steady-state error (median 0.0065°) and settled within 1° in 3.6 s. Standard PPO showed little variation across seeds (4.7 ± 0.4 s; 0.40 ± 0.02° residual error); its residual error arises because its observation contains no memory of past error. Adding the integral of the error to the observation lowered the residual error on average (0.17 ± 0.15°) and, in the best seed, matched the PID (0.0056°), but results varied strongly between seeds and settling was slower (8.5 ± 2.5 s). The RL agents used on average about 16–26% less control torque than PD/PID, partly because their reward penalizes effort. A first single-seed run had suggested a twelve-fold error reduction for PPO with the integral; the multi-seed study showed this was a favorable draw, illustrating the importance of strong baselines and repeated training runs.

1. Introduction

Attitude control keeps a spacecraft pointed in a desired direction despite disturbances such as solar radiation pressure, magnetic torques, and residual momentum. Quaternion-feedback PD/PID controllers are the long-standing baseline for this task [4, 5]. Recently, deep RL methods such as PPO [1] have been proposed as alternatives that learn a control policy from interaction rather than from an analytical design. A fair assessment of such methods requires a carefully tuned classical baseline, a held-out evaluation set, several independent training runs, and metrics beyond a single trajectory.

This work asks a simple question: *in a controlled simulation, does a PPO policy outperform a properly tuned PID controller for rest-to-rest attitude maneuvers under saturation and an unknown constant disturbance?* A secondary hypothesis was that a memoryless PPO policy cannot cancel a constant disturbance, and that giving it the integral of the attitude error, the same information a PID integrator uses, would close that gap. The contributions are:

- a reproducible comparison protocol (separate tuning/training and test scenarios, four controllers, four metrics, five training seeds per RL variant);
- an ablation testing whether an error-integral observation gives PPO the missing “memory”, and how reliably;
- open-source code, trained models and raw results, so that every number in this report can be regenerated.

2. Methods

2.1 Spacecraft model

The spacecraft is a rigid body with state $x = [q, \omega]$, where q is the unit attitude quaternion (scalar first) and ω the body-frame angular velocity. The dynamics are

$$\mathrm{d}q/\mathrm{d}t = \tfrac{1}{2}\, q \otimes [0, \omega], \quad J\,\mathrm{d}\omega/\mathrm{d}t = \tau + d - \omega \times (J\omega),$$

with inertia $J = \mathrm{diag}(0.05, 0.06, 0.04)$ kg·m², commanded torque τ limited to ±0.05 N·m per axis, and an unknown constant disturbance torque d. Equations are integrated with a fourth-order Runge–Kutta scheme (step 0.02 s, quaternion renormalized every step); the controller acts every 0.1 s with a zero-order hold. The simulator was verified against two analytic checks: with zero torque, the angular momentum norm is conserved to about 10^{-17}, and with a constant torque about one axis from rest, the angular velocity matches $\tau/J \cdot t$ to machine precision.

2.2 PD and PID controllers

Let $q_e = q_\mathrm{d}^{-1} \otimes q$ be the error quaternion and e its vector part, multiplied by the sign of its scalar part so that the shorter rotation is always chosen. The control law is

$$\tau = -K_p\, e - K_d\, \omega - K_i \int e\, \mathrm{d}t,$$

followed by saturation. The PD controller is the case $K_i = 0$. To avoid integral windup during large slews, the integrator only accumulates when $\|e\| < 0.1$ (an error angle below about 11.5°) and the commanded torque is not saturated, and its state is clamped [5]. Gains were chosen by grid search over $K_p \in \{0.2, 0.4, 0.8, 1.6, 3.2\}$, $K_d \in \{0.1, 0.2, 0.4\}$ and, for the PID, $K_i \in \{0.01, 0.03, 0.06\}$, minimizing the mean of (settling time + 20 × final error) on 10 tuning scenarios. The selected gains are PD: $(K_p, K_d) = (1.6, 0.4)$ and PID: $(K_p, K_d, K_i) = (0.8, 0.2, 0.06)$.

2.3 PPO agents

The simulator was wrapped as a Gymnasium [3] environment. The observation is the error vector e and the scaled angular velocity (6 values). The action is a 3-axis torque in $[-1, 1]$, scaled to the 0.05 N·m limit. The reward is dense:

$$r = -(\|e\| + 0.2\|\omega\| + 0.05\|u\|^2) + 0.1 \cdot [\|e\| < 0.0087],$$

where the bracket rewards being within about 1° of the target. Each 40 s episode starts from a random attitude (rotation of 20°–150° about a random axis), a random initial spin (±0.05 rad/s per axis) and a random constant disturbance (±0.003 N·m per axis); the target attitude is fixed. The **PPO+I** variant adds three observations, the scaled integral of e, accumulated under the same rule and clamp as the PID integrator; nothing else differs, so the comparison PPO vs. PPO+I isolates the effect of this memory. Both variants use Stable-Baselines3 [2] PPO with the default MLP policy, learning rate 3×10^{-4}, 2048 steps per update, batch size 64, discount 0.99 and 300,000 environment steps (about 20 minutes on a laptop CPU). **Each variant was trained five times with independent random seeds (0–4)**, and every trained policy is evaluated deterministically.

2.4 Evaluation protocol

All controllers use the same simulator, torque limit and control period. Tuning used scenario seeds 0–9; testing used 100 different scenarios (seeds 1000–1099) drawn from the same distribution as the RL training episodes, each simulated for 60 s. Metrics: *settling time* (last instant the attitude error exceeds 1°), *final error* (mean attitude error over the last 5 s), *RMS torque* (√ of the time-mean of $\|\tau\|^2$) and *torque jerk* (mean $\|\tau_k - \tau_{k-1}\|$ per control step). *Uncertainty.* For the deterministic PD and PID controllers, the 95% bootstrap confidence interval of the median over the 100 scenarios is given. For each PPO variant, the median over scenarios is computed per trained seed, and the mean ± standard deviation (and range) across the five seeds is reported. Finally, a paired comparison counts, scenario by scenario, how often the PID outperforms each RL model. All comparisons are descriptive: no statistical significance tests were performed and no claim of statistical significance is made.

3. Results

Table 1 and Figure 1 summarize the four controllers over the 100 test scenarios.

Controller	Settling time to 1° (s)	Final error (°)	RMS torque (N·m)	Torque jerk ($\times 10^{-3}$)
PD (tuned)	3.50 [3.10–3.80]	0.205 [0.190–0.225]	0.0143	0.71
PID, anti-windup (tuned)	3.60 [3.10–4.10]	**0.0065** [0.0061–0.0072]	0.0139	0.64
PPO (5 seeds)	4.67 ± 0.41 (4.20–5.25)	0.401 ± 0.023 (0.369–0.424)	0.0117 ± 0.0024	0.61 ± 0.25
PPO+I (5 seeds)	8.45 ± 2.47 (5.80–12.35)	0.168 ± 0.153 (0.0056–0.343)	**0.0106 ± 0.0022**	**0.49 ± 0.21**

Table 1. Results on 100 unseen scenarios (60 s each, control period 0.1 s, torque limit 0.05 N·m per axis). Settling time and final error: PD and PID show the median over the scenarios [95% bootstrap confidence interval]; PPO and PPO+I show the mean ± standard deviation over five training seeds of each seed's median, with the range of the seed medians in parentheses. RMS torque and torque jerk: means over the 100 scenarios; no confidence interval was computed for PD and PID, and for PPO and PPO+I the ± is the standard deviation across seeds of each seed's scenario mean. Best value in bold for final error, RMS torque and jerk. The comparison is descriptive; no significance tests were performed.

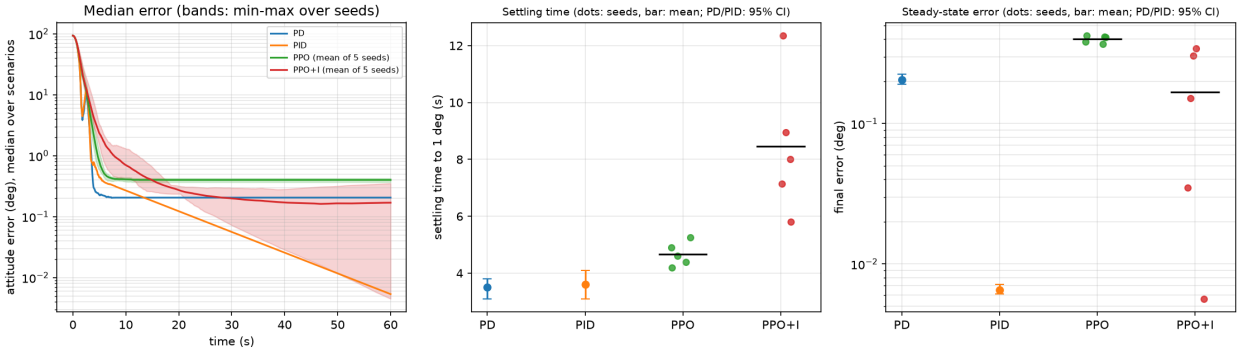


Figure 1. Left: median attitude error over the 100 scenarios (log scale). For the RL variants, the line is the mean over five seeds of each seed's median curve and the band spans the minimum and maximum seed. Centre and right: settling time and final error; each dot is one training seed (bar: mean). For PD and PID the marker is the median with its 95% confidence interval.

Classical controllers. PD and PID have nearly identical settling times (3.5 s and 3.6 s; the confidence intervals overlap), but only the PID removes the effect of the constant disturbance: its final error (0.0065°) is about 30 times smaller than the PD's (0.205°).

Standard PPO showed little variation across the five training seeds. It settles in 4.7 ± 0.4 s and plateaus at 0.40 ± 0.02°, about twice the PD's error and 60 times the PID's. Even the

fastest PPO seed (4.2 s) was slower than the upper ends of the confidence intervals of both classical controllers (3.8 s and 4.1 s); this is a descriptive observation, not a significance test.

PPO+I. Giving the policy the error integral reduced the mean residual error from 0.40° to 0.17° (about 2.4 times), but with large seed-to-seed variation: from 0.0056° (matching the PID) to 0.343° (close to plain PPO). Its settling time was also longer and more variable (8.5 ± 2.5 s; range 5.8–12.4 s). A first single-seed run (seed 0) had suggested a twelve-fold error reduction (0.416° to 0.035°); the five-seed study shows that this was a favorable draw.

Control effort. On average the RL agents used less torque: RMS 0.0117 ± 0.0024 N·m (PPO) and 0.0106 ± 0.0022 N·m (PPO+I), versus 0.0143 (PD) and 0.0139 (PID), i.e. about 16–18% and 24–26% less. The advantage is not uniform: one PPO seed used more torque (0.0159 N·m) than either classical controller. **Paired comparison.** Scenario by scenario (averaged over seeds), the PID settled strictly faster than PPO in 82% and than PPO+I in 98% of the scenarios, and had a lower final error in 100% and 87%, respectively.

Lesson on baselines. An early test on a single 90° slew, using a lightly hand-tuned PID, suggested that PPO was about five times faster (4.1 s vs. 20.4 s to reach 1°). Systematic tuning of the classical controllers and testing on 100 scenarios reversed this conclusion, which is why the protocol separates tuning from testing.

4. Discussion

Why the PID wins here. The task is a smooth, nearly linear regulation problem for which a feedback law with an integrator is close to ideal; the constant disturbance is exactly what the integral term is designed to reject. A memoryless policy cannot infer an unknown constant torque from a single observation, which explains PPO’s steady offset. Giving the policy the error integral provides the necessary information and, on average, the offset dropped, so the information is used; but learning to use it reliably is harder.

Why PPO+I is slow and variable. The cause was not isolated in this study. The large spread between seeds (one seed matches the PID, while some stay close to the plain-PPO level) points to an optimization difficulty: a policy with memory must learn a slow integral action from a reward that gives little incentive to correct small errors. Plausible remedies are longer training and a reward that penalizes small errors more strongly; both are untested.

Control effort. The lower average torque of the RL agents should not be read as an intrinsic advantage: their reward explicitly penalizes effort, while the PD/PID gains were tuned only for settling time and final error. A fair multi-objective comparison would add an effort term to the classical tuning objective or compare Pareto fronts.

Why several seeds matter. A single-seed comparison would have reported a twelve-fold improvement for PPO+I; the five-seed mean is 2.4-fold with a spread of almost two orders of magnitude between seeds. RL results should therefore be reported with their seed-to-seed variability.

Limitations.

- Simulation only: no sensor noise, actuator dynamics, flexible modes or delays; the inertia is known and fixed.
- Only constant disturbances were considered; the test distribution matches the RL training distribution, so generalization to other conditions was not assessed.
- Five training seeds per RL variant is a small sample, so the standard deviations are themselves uncertain. The results are reported descriptively; no significance tests were performed and no claim of statistical significance is made.

- The classical gains come from a small grid search, and the PPO hyperparameters and reward weights were not tuned.

5. Future Work

- More seeds, longer training and reward tuning (e.g., a stronger incentive for small errors), especially for PPO+I.
- Robustness tests: sensor noise, time-varying disturbances, inertia uncertainty and reaction-wheel dynamics, with domain randomization during training.
- Stronger classical baselines (LQR, model-predictive control) and multi-objective (effort vs. accuracy) comparisons.

6. Conclusion

In this simulation study, a properly tuned PID controller outperformed PPO in settling speed and steady-state accuracy under saturation and a constant disturbance, while the PPO agents used less control torque on average. Adding the error integral to the PPO observation lowered the steady-state error on average and, in one of five seeds, matched the PID, but not reliably. The more general result is methodological: comparisons between RL and classical controllers should use carefully tuned baselines, held-out scenarios and multiple training seeds; here a single favorable seed overstated the benefit of the integral observation about five-fold (12× vs. 2.4×). All code, trained models and raw results are available in the open-source repository at github.com/yasinkh81b-ui/ppo-vs-pid-spacecraft-attitude.

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References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.

[2] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann. Stable-Baselines3: Reliable Reinforcement Learning Implementations. Journal of Machine Learning Research 22(268):1–8, 2021.

[3] M. Towers et al. Gymnasium: A Standard Interface for Reinforcement Learning Environments. arXiv:2407.17032, 2024.

[4] B. Wie, P. M. Barba. Quaternion Feedback for Spacecraft Large Angle Maneuvers. Journal of Guidance, Control, and Dynamics 8(3):360–365, 1985.

[5] K. J. Åström, T. Hägglund. Advanced PID Control. ISA, 2006.